

Patrick Pynadath

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ABOUT ME

Bio	I am a PhD student at Purdue University, studying Machine Learning under the guidance of Professor Ruqi Zhang. I am broadly interested in probabilistic machine learning methodologies, with a current focus on diffusion language models (DLMs)
Interests	Diffusion Language Models, MCMC, LLMs, Generative Modeling, Controlled Generation
Frameworks	PyTorch, PyTorch Lightning, Hydra, Slurm, TensorFlow, NumPy, Matplotlib, BeautifulSoup, Hugging Face Transformers

EDUCATION

Purdue University , <i>PhD, Computer Science</i>	Aug. 2023 – Present
Northwestern University , <i>BA Mathematics, MS Computer Science</i>	Sept. 2018 – Dec. 2022

EXPERIENCE

Student Researcher, *Google DeepMind* June 2026 – Sept. 2026

- Researched faster sampling algorithms for diffusion language models with the DiffusionGemma team, improving inference efficiency over state-of-the-art baselines.
- Designed and built an end-to-end evaluation harness from scratch, covering model generation, scoring, sandboxing for code evaluation, LLM autoraters, sampling trace analysis, and results visualization.
- Developed a novel sampling method with substantial efficiency gains over the prior approach (under embargo, publication pending).

Research Assistant, *Purdue University* Jan. 2024 – June 2026

- Conduct research in probabilistic machine learning, with work spanning diffusion models, discrete MCMC (Markov Chain Monte Carlo) methods, and variational inference for large language models.
- Built large-scale experimental pipelines using PyTorch, PyTorch Lightning, Hydra, and Slurm for training and evaluating language and generative models.

PUBLICATIONS

SEED: Self-Speculative Decoding via Implicit Encoder–Decoder Preprint 2026

Hankun Lin, *Patrick Pynadath*, Ruqi Zhang

In progress

Sticky Jump Diffusions: A Unifying View of Masked, Continuous, and Hybrid Diffusion Preprint 2026

Pascal Jutras-Dubé, *Patrick Pynadath*, Jeremy Lu, Yuan Gao, Ruqi Zhang

arXiv:2607.10951

Simplex Relaxation for Discrete Diffusion Preprint 2026

Jinya Sakurai, *Patrick Pynadath*, Satoshi Hayakawa, Jaehong Yoon, Xulei Yang, Nancy F. Chen, Xun Xu

arXiv:2608.10615

Generative Frontiers: Why Evaluation Matters for Diffusion Language Models Technical Note 2026

Patrick Pynadath, Jiaxin Shi, Ruqi Zhang

arXiv:2604.02718

Why Any-Order Autoregressive Models Need Two-Stream Attention: A Structural-Semantic Tradeoff
Patrick Pynadath, Ruqi Zhang
[arXiv:2602.16092](#)

Technical Note 2026

CANDI: Hybrid Discrete-Continuous Diffusion Models
Patrick Pynadath, Jiaxin Shi, Ruqi Zhang
[arXiv:2510.22510](#)

ICML 2026

VERA: Variational Inference Framework for Jailbreaking Large Language Models
Anamika Lochab*, Lu Yan*, *Patrick Pynadath**, Xiangyu Zhang, Ruqi Zhang
[arXiv:2506.22666](#) • * *equal contribution*

NeurIPS 2025

Controlled LLM Decoding via Discrete Auto-regressive Biasing
Patrick Pynadath, Ruqi Zhang
[arXiv:2502.03685](#)

ICLR 2025

Gradient-based Discrete Sampling with Automatic Cyclical Scheduling
Patrick Pynadath, Riddhiman Bhattacharya, Arun Hariharan, Ruqi Zhang
[arXiv:2402.17699](#)

NeurIPS 2024